QUBO Formulation: LLM Cascade Routing Problem

LLM Cascade Routing Problem

Problem Statement. Given a set of N user requests and a cascade of K available language model tiers, the objective is to assign each request to exactly one model tier. Each assignment (r, m) incurs an inference cost $c_{r,m}$ and yields a predicted quality score $q_{r,m}$. Each request r has a minimum quality requirement τ_r. The task is to choose exactly one model for every request such that the total inference cost is minimized while satisfying all quality requirements.

Decision Variables. For each request $r \in \{0, \dots, N-1\}$ and each model tier $m \in \{0, \dots, K-1\}$, we introduce a binary decision variable $x_{r,m} \in \{0, 1\}$. The variable $x_{r,m} = 1$ if and only if request r is routed to model tier m, and $x_{r,m} = 0$ otherwise. The total number of variables is $N \times K$.

QUBO Derivation. We construct the QUBO Hamiltonian $H(x)$ by encoding the objective function and constraints into a single quadratic pseudo-boolean function.

Step 1 – Objective Function. The original minimization objective is

$$\min \sum_{r=0}^{N-1} \sum_{m=0}^{K-1} c_{r,m} x_{r,m}. \tag{1}$$

Quality requirements are enforced by modifying the linear coefficients. Define a validity parameter $\text{valid}_{r,m} \in \{0, 1\}$, where $\text{valid}_{r,m} = 1$ if $q_{r,m} \geq \tau_r$ and $\text{valid}_{r,m} = 0$ otherwise. We incorporate a large penalty weight P into the linear term to heavily penalize invalid assignments:

$$\tilde{c}_{r,m} = c_{r,m} + P(1 - \text{valid}_{r,m}). \tag{2}$$

The objective becomes $\min \sum_{r=0}^{N-1} \sum_{m=0}^{K-1} \tilde{c}_{r,m} x_{r,m}$.

Step 2 – Constraints. Each request must be routed to exactly one model tier. This is expressed as a set of equality constraints:

$$\sum_{m=0}^{K-1} x_{r,m} = 1, \quad \forall r \in \{0, \dots, N-1\}. \tag{3}$$

Step 3 – Penalty Term Conversion. Each equality constraint is converted to a quadratic penalty term with weight P:

$$\mathcal{P}_r = P\left(\sum_{m=0}^{K-1} x_{r,m} - 1\right)^2. \tag{4}$$

Expanding the square and applying the idempotent property $x_{r,m}^2 = x_{r,m}$ for binary variables yields:

$$\begin{aligned}\left(\sum_{m=0}^{K-1} x_{r,m} - 1\right)^2 &= \sum_{m=0}^{K-1} x_{r,m}^2 + \sum_{m=0}^{K-1}\sum_{\substack{m'=0\\ m'\neq m}}^{K-1} x_{r,m}x_{r,m'} - 2\sum_{m=0}^{K-1} x_{r,m} + 1 \qquad (5)\\ &= 1 - \sum_{m=0}^{K-1} x_{r,m} + \sum_{m=0}^{K-1}\sum_{\substack{m'=0\\ m'\neq m}}^{K-1} x_{r,m}x_{r,m'}. \qquad (6)\end{aligned}$$

Thus, the penalty contribution for request r is:

$$\mathcal{P}_r = P - P\sum_{m=0}^{K-1} x_{r,m} + P\sum_{m=0}^{K-1}\sum_{\substack{m'=0\\ m'\neq m}}^{K-1} x_{r,m}x_{r,m'}. \tag{7}$$

Step 4 – Combined Hamiltonian. Summing the objective and all penalty terms gives the full QUBO Hamiltonian:

$$H(x) = \sum_{r=0}^{N-1}\sum_{m=0}^{K-1} \tilde{c}_{r,m}x_{r,m} + P\sum_{r=0}^{N-1}\left(1 - \sum_{m=0}^{K-1} x_{r,m} + \sum_{m=0}^{K-1}\sum_{\substack{m'=0\\ m'\neq m}}^{K-1} x_{r,m}x_{r,m'}\right). \tag{8}$$

Step 5 – Q Matrix Entries and Offset. Mapping each pair (r, m) to a single index $i = rK+m$, the Hamiltonian takes the standard QUBO form $H(x) = \sum_i Q_{i,i}x_i + \sum_{i<j} Q_{i,j}x_ix_j + c$. The coefficients are read off as:

$$\begin{aligned} Q_{i,i} &= \tilde{c}_{r,m} - P, &(9)\\ Q_{i,j} &= 2P \quad \text{if } r = r' \text{ and } m \neq m', &(10)\\ Q_{i,j} &= 0 \quad \text{if } r \neq r', &(11)\\ c &= NP. &(12)\end{aligned}$$

Penalty Weight Justification. The penalty weight P must strictly exceed the maximum possible difference in valid routing costs to guarantee constraint satisfaction. Specifically, choosing $P > \max_{r,m} c_{r,m}$ ensures that the penalty P incurred by selecting zero or two models for any request (which adds P to the energy) always outweighs any potential cost savings from an invalid assignment. This bound guarantees that the energy of any infeasible configuration is strictly higher than the energy of the best feasible configuration.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{NK}$ recovers the optimal assignment because (i) any feasible solution satisfies $\sum_m x_{r,m} = 1$ for all r, yielding a penalty of zero and an energy equal to the total inference cost; (ii) any infeasible solution violates at least one one-hot constraint, incurring a penalty of at least P. Since P is chosen to dominate the maximum possible valid cost, every infeasible solution has strictly higher energy than the optimal feasible solution. Consequently, the global minimum of $H(x)$ must be feasible and correspond to the minimum-cost valid routing.

Illustrative Example. Consider a concrete instance with $N = 3$ requests and $K = 2$ model tiers. The variables are mapped to indices $0, \ldots, 5$ via $i = 2r + m$. Let the inference costs be $c_{0,0} = 10, c_{0,1} = 20, c_{1,0} = 15, c_{1,1} = 25, c_{2,0} = 12, c_{2,1} = 18$. Assume all assignments satisfy quality thresholds, so $\text{valid}_{r,m} = 1$ and $\tilde{c}_{r,m} = c_{r,m}$. Set the penalty weight $P = 100$.

The concrete linear terms are $10x_0, 20x_1, 15x_2, 25x_3, 12x_4, 18x_5$. The penalty terms for each request are:

$$\mathcal{P}_0 = 100(1 - x_0 - x_1 + 2x_0x_1), \tag{13}$$

$$\mathcal{P}_1 = 100(1 - x_2 - x_3 + 2x_2x_3), \tag{14}$$

$$\mathcal{P}_2 = 100(1 - x_4 - x_5 + 2x_4x_5). \tag{15}$$

The resulting Hamiltonian is:

$$H(x) = 10x_0 + 20x_1 + 15x_2 + 25x_3 + 12x_4 + 18x_5 \tag{16}$$

$$+ 100(1 - x_0 - x_1 + 2x_0x_1) + 100(1 - x_2 - x_3 + 2x_2x_3) + 100(1 - x_4 - x_5 + 2x_4x_5). \tag{17}$$

Collecting terms, the Q matrix diagonal entries are $Q_{0,0} = -90, Q_{1,1} = -80, Q_{2,2} = -85, Q_{3,3} = -75, Q_{4,4} = -88, Q_{5,5} = -82$. Off-diagonal entries are $Q_{0,1} = Q_{2,3} = Q_{4,5} = 200$, with all others zero. The constant offset is $c = 300$.

Evaluating feasible assignments (where exactly one variable per request pair is 1), the assignment $x_{0,0} = 1, x_{1,0} = 1, x_{2,0} = 1$ (i.e., $x_0 = 1, x_2 = 1, x_4 = 1$, all others 0) yields energy $H = 10 + 15 + 12 + 0 = 37$. Any assignment routing to model 1 or violating the one-hot constraint yields energy ≥ 100. Thus, the optimal solution is $x_0 = 1, x_2 = 1, x_4 = 1$ with minimum total cost 37.